

1. Administrative & Study Identification

Full Study Title: A Study to Assess the Efficacy, Safety and Tolerability of Different Doses of AZD0780 in Patients With Dyslipidemia **Short Title/Acronym:** PURSUIT **Protocol Number:** D7960C00006 **NCT Number:** NCT06173570 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD0780 (Oral PCSK9 Inhibitor) **Phase of Development:** Phase II **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To measure the effect of different daily doses of AZD0780 on Low-Density Lipoprotein (LDL-C) levels (measured by percent change from baseline to Week 12) compared with placebo in participants with dyslipidemia. **Secondary Objectives:** To investigate the effect of AZD0780 versus placebo on other lipid parameters and inflammatory markers, assess blood concentrations of AZD0780 at specific timepoints (pharmacokinetics), and evaluate its safety and tolerability profile.

2.2 Background & Rationale Most patients at high risk for cardiovascular events do not achieve lipid goals advocated by American College of Cardiology/American Heart Association (ACC/AHA) guidelines despite the wide availability of lipid-lowering therapies (e.g., statins). While subcutaneously administered PCSK9 inhibitors have been shown to significantly lower LDL-C, no oral PCSK9 inhibitors are currently available. This study aims to evaluate AZD0780, a novel, oral, small-molecule inhibitor of PCSK9, which could provide an effective once-daily alternative for eligible patients who are unable or unwilling to take injectable drugs.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional (Randomized Controlled Trial) **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1:1:1 (AZD0780 1 mg, 3 mg, 10 mg, 30 mg, or matching placebo)

3.2 Planned Enrollment & Duration **Target N\$:** 428 patients **Number of Sites:** Multicenter **Estimated Study Start:** [28-Jan-2024] **Estimated Primary Completion:** [03-Jun-2025]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Males, and females of non-childbearing potential, aged 18 to 75 years, inclusive. 2. Fasting low-density lipoprotein cholesterol (LDL-C) ≥ 70 mg/dL (1.8 mmol/L) and < 190 mg/dL (4.9 mmol/L) at screening. 3. Fasting triglycerides < 400 mg/dL at screening, and currently receiving moderate or high-intensity statin therapy for ≥ 2 months prior to screening (with or without ezetimibe).

4.2 Exclusion Criteria 1. Poorly controlled type 2 diabetes mellitus, defined as hemoglobin A1c (HbA1c) $> 10\%$ at screening. 2. Acute ischemic cardiovascular event in the last 12 months, or Heart failure with New York Heart Association (NYHA) Class III-IV. 3. Malignancy within the last 10 years, recipient of any major organ transplant, or LDL/plasma apheresis within 12 months prior to randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: AZD0780 (1 mg, 3 mg, 10 mg, or 30 mg) or matching placebo administered once daily. **Route of Administration:** Oral **Duration of Treatment:** 12 weeks

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of moderate- or high-intensity statins, with or without ezetimibe. **Prohibited:** Planned medication or statin dose changes during study participation.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, Fasting Lipid Profile, HbA1c
Baseline	Day 0	Randomization, First Dose, Vitals, ECG	End of Study
Week 12	Final Efficacy Labs (LDL-C), Vital Signs, ECG, Safety Labs, IP Accountability	Follow-Up	Post-Treatment Off-treatment safety follow-up and monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percent change of LDL-C from baseline to week 12. **Measurement Method:** Central Laboratory Analysis of fasting lipid panels.

7.2 Secondary & Exploratory Endpoints * Effect on other lipid parameters and inflammatory markers from baseline. * Concentration of AZD0780 in blood at specific timepoints (Pharmacokinetics). * Number of treatment-emergent adverse events

(TEAEs), vital signs, electrocardiograms (ECGs), and safety laboratory assessments.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of adverse events, safety labs, vital signs, and ECGs routinely throughout the 12-week treatment and follow-up phase. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Internal/External safety monitoring committee details managed per AstraZeneca protocol.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Full Analysis Set for primary efficacy evaluations; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N=428$ sufficiently powered to detect a statistically significant placebo-corrected difference in least squares mean percent change of LDL-C across 4 dose cohorts vs. placebo. **Handling of Missing Data:** Standard imputation (e.g., Mixed Model for Repeated Measures [MMRM] or Multiple Imputation) for missing Week 12 lipid values.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits per AstraZeneca clinical operations standards. **Audit Trail:** eCRF locking and data clarification forms generated by site monitors.

11. Study Identifiers & Governance

Primary ID: D7960C00006 **Registry Number:** NCT06173570 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PURSUIT **Official Regulatory Title:** A Study to Assess the Efficacy, Safety and Tolerability of Different Doses of AZD0780 in Patients With Dyslipidemia

12. Clinical Framework (Dyslipidemia Specific)

Primary Condition: Dyslipidemia / Hypercholesterolemia **Diagnosis Threshold:** Fasting LDL-C ≥ 70 mg/dL and < 190 mg/dL **Medical Methodology:** Oral PCSK9 inhibition added to background statin therapy **Optimization Target:** ACC/AHA guideline LDL-C goal for high-risk patients

13. Study Procedures & Methodology

Management Pathway: Standard lipid-lowering therapy + AZD0780
Education Protocol (HCP): Protocol specific dosing, compliance, and safety reporting training
Education Protocol (Patient): Daily oral drug compliance and fasting lab requirements
Quality Audit Mechanism: Clinical site monitoring and data verification

14. Observation & Follow-Up Schedule

Total Duration: 12 weeks of active treatment + off-treatment follow-up
Onsite Visit Cadence: Screening, Baseline, Week 12, Follow-up
Remote Monitoring Frequency: As per protocol requirements
Checklist Review Interval: At every protocol-mandated visit

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				